CIE AS PHYSICS (9702) • QUESTION PAPER

1.2 SI Units

Syllabus Section: 1. Physical Quantities & Units **Paper Type:** Multiple Choice (Paper 1) **Total:** 22 Questions (22 Marks)

Question 1 Easy [1 mark]

Which of the following is an SI base unit?

A Current

B Gram

C Kelvin

D Volt

Question 2 Easy [1 mark]

Five energies are listed below.

5kJ

5mJ

5MJ

5nJ

Starting with the smallest first, what is the order of increasing magnitude of these energies?

A 5kJ → 5mJ → 5MJ → 5nJ

B 5nJ → 5kJ → 5MJ → 5mJ

C 5nJ → 5mJ → 5kJ → 5MJ

D 5mJ → 5nJ → 5kJ → 5MJ

Question 3 Easy [1 mark]

Which product-pair of metric prefixes has the greatest magnitude?

A pico × mega

B nano × kilo

C micro × giga

D milli × tera

Question 4 Easy [1 mark]

Which statement includes a correct unit?

A Energy = 7.8 N s

B Force = 3.8 N s

C Momentum = 6.2 N s

D Torque = 4.7 N s

Question 5 Easy [1 mark]

Which quantity can be measured in electronvolts (eV)?

A Electric charge

B Electric potential

C Energy

D Power

Question 6 Easy [1 mark]

Which of the following is not a SI base unit?

A kilogram

B metre

C second

D volt

Question 7 Easy [1 mark]

Which product pair of metric prefixes has the smallest magnitude?

A nano × mega

B tera × pico

C centi × kilo

D micro × giga

Question 8 Easy [1 mark]

Which unit is **not** an SI base unit?

A A

B K

C mol

D N

Question 9 Medium [1 mark]

The average kinetic energy E of a gas molecule is given by the equation

$$E = \frac{3}{2}kT$$

where T is the thermodynamic (kelvin) temperature.

What are the SI base units of k?

A $\text{kg}^{-1}\ \text{m}^{-1}\ \text{s}^{2}\ \text{K}$

B $\text{kg}^{-1}\ \text{m}^{-2}\ \text{s}^{2}\ \text{K}$

C $\text{kg}\ \text{m}\ \text{s}^{-2}\ \text{K}^{-1}$

D $\text{kg}\ \text{m}^{2}\ \text{s}^{-2}\ \text{K}^{-1}$

Question 10 Medium [1 mark]

The units of all physical quantities can be expressed in terms of SI base units.

Which pair contains quantities with the same base units?

A Force and momentum

B Pressure and Young modulus

C Power and kinetic energy

D Mass and weight

Question 11 Medium [1 mark]

The table contains some quantities, together with their symbols and units.

Quantity	Symbol	Unit
Gravitational field strength	g	N kg^{-1}
Density of liquid	ρ	kg m^{-3}
Vertical height	h	m
Volume of part of liquid	V	m^3

Which expression has the units of energy?

A $\rho g^2 h$

B $\frac{\rho h V}{g}$

C $\frac{\rho g}{h V}$

D $g \rho h V$

Question 12 Medium [1 mark]

What is the ratio $\frac{10^{-3}\ \text{THz}}{10^{3}\ \text{kHz}}$?

A 10^{-9}

B 10^{-6}

C 10^{0}

D 10^{3}

Question 13 Medium [1 mark]

Which of the following correctly expresses the volt in terms of SI base units?

A AΩ

B WA^{-1}

C $\text{kg}\ \text{m}^2\ \text{s}^{-1}\ \text{A}^{-1}$

D $\text{kg}\ \text{m}^2\ \text{s}^{-3}\ \text{A}^{-1}$

Question 14 Medium [1 mark]

The equation relating pressure and density is $P = \rho g h$

How can both sides of this equation be written in terms of base units?

A $\mathrm{N\ m^{-1} = kg\ m^{-3}\ m\ s^{-1}\ m}$

B $\mathrm{N\ m^{-2} = kg\ m^{-2}\ m\ s^{-2}\ m}$

C $\mathrm{kg\ m^{-1}\ s^{-2} = kg\ m^{-3}\ m\ s^{-2}\ m}$

D $\mathrm{kg\ m^{-1}\ s^{-1} = kg\ m^{-1}\ m\ s^{-2}\ m}$

Question 15 Medium [1 mark]

What is the ratio $\frac{10^{2}\ \mathrm{THz}}{10^{-4}\ \mathrm{pHz}}$?

A 10^{-30}

B 10^{-2}

C 10^{2}

D 10^{30}

Question 16 Hard [1 mark]

The speed *v* of a liquid leaving a tube depends on the change in pressure ΔP and the density ρ of the liquid. The speed is given by the equation

$$v = k\frac{\Delta P}{\rho}^{n}$$

where *k* is a constant that has no units.

What is the value of *n*?

A $\frac{1}{2}$

B 1

C $\frac{3}{2}$

D 2

Question 17 Hard [1 mark]

The frictional force *F* on a sphere falling through a fluid is given by the formula

$$F = 6\pi r\eta v$$

where *r* is the radius of the sphere, η is a constant relating to the fluid and *v* is the velocity of the sphere.

What are the units of η ?

A $\mathrm{kg\ m\ s^{-1}}$

B $\mathrm{kg\ m^{-1}\ s^{-1}}$

C $\mathrm{kg\ m\ s^{-3}}$

D $\mathrm{kg\ m^{3}\ s^{-3}}$

Question 18 Hard [1 mark]

When a constant braking force is applied to a vehicle moving at speed *v*, the distance *d* moved by the vehicle as it comes to rest is given by the expression

$$d = kv^2$$

where *k* is a constant.

When *d* is measured in metres and *v* is measured in metres per second, the constant has a value of k_1.

What is the value of the constant *k* when the distance is measured in metres, and the speed is measured in kilometres per hour?

A $0.077k_1$

B $0.278k_1$

C $3.60k_1$

D $13.0k_1$

Question 19 Hard [1 mark]

The drag coefficient C_d is a number with no units. It is used to compare the drag on different cars at different speeds. It is given by the equation

$$C_d = \frac{2F}{\rho v^n A}$$

where F is the drag force on the car, ρ is the density of the air, A is the cross-sectional area of the car and v is the speed of the car.

What is the value of n?

A 1

B 2

C 3

D 4

Question 20 Hard [1 mark]

Which formula could be correct for the speed v of ocean waves in terms of the density ρ of sea-water, the acceleration of free fall g, the height h of the wave and the wavelength λ?

A $v = \sqrt{\rho g h}$

B $v = \sqrt{\frac{g}{h}}$

C $v = \sqrt{g\lambda}$

D $v = \sqrt{\frac{g}{\rho}}$

Question 21 Hard [1 mark]

The equation for the terminal velocity of a sphere of radius *r* falling under acceleration due to gravity, *g*, through a liquid with density ρ and viscosity η, measured in kg m^{-1} s^{-1} is:

$$v_t = \frac{2}{9}\frac{gr^n\rho}{\eta}$$

What is the value of *n*?

A 0.5

B 1

C 2

D 3

Question 22 Hard [1 mark]

The current I through a semiconductor diode is given by:

$$I = I_0\, e^{\frac{-qV}{nT}}$$

where I_0 is the initial current, q is the charge of an electron, V is the applied potential difference, T is the temperature, and n is a constant.

What is the unit of n when expressed in SI base units?

A kg m^2 s^{-2} K^{-1}

B kg m^2 s^{-4} K^{-1}

C kg^{-1} m^{-2} s^2 K

D kg^{-1} m^{-2} s^4 K